SURVEYING & LEVELLING · NOTE SL10

Levelling — Terms, Bench Marks, Staff Readings and Instruments

Course topics: SWCE1.4 L10

What levelling does

Levelling measures in the **vertical plane**: it finds the relative heights (elevations) of points on the earth's surface, directly, with a level and a graduated rod.

Farm uses: foundations of farm buildings; grade lines for irrigation and drainage channels; land levelling; contour lines for strip cropping, contour bunds and terraces; underground pipeline water distribution systems.

Basic terms

Term	Meaning
Level surface (plane surface)	A surface perpendicular to the direction of gravity at every point
Level line	A line lying on a level surface
Horizontal plane	A plane perpendicular to the plumb line at a point — tangential to the level surface there
Horizontal line	A line lying in the horizontal plane
Vertical line	The line in which a plumb bob hangs; perpendicular to the level surface
Vertical plane	A plane containing a vertical line
Datum (datum surface, datum plane)	The reference to which all elevations are referred; elevation zero. Mean Sea Level (MSL) is the standard datum; minor surveys may assume one
Elevation / reduced level (RL)	Vertical distance of a point above (+) or below (-) the datum
Difference in elevation	Vertical distance between the level surfaces through two points

Bench marks

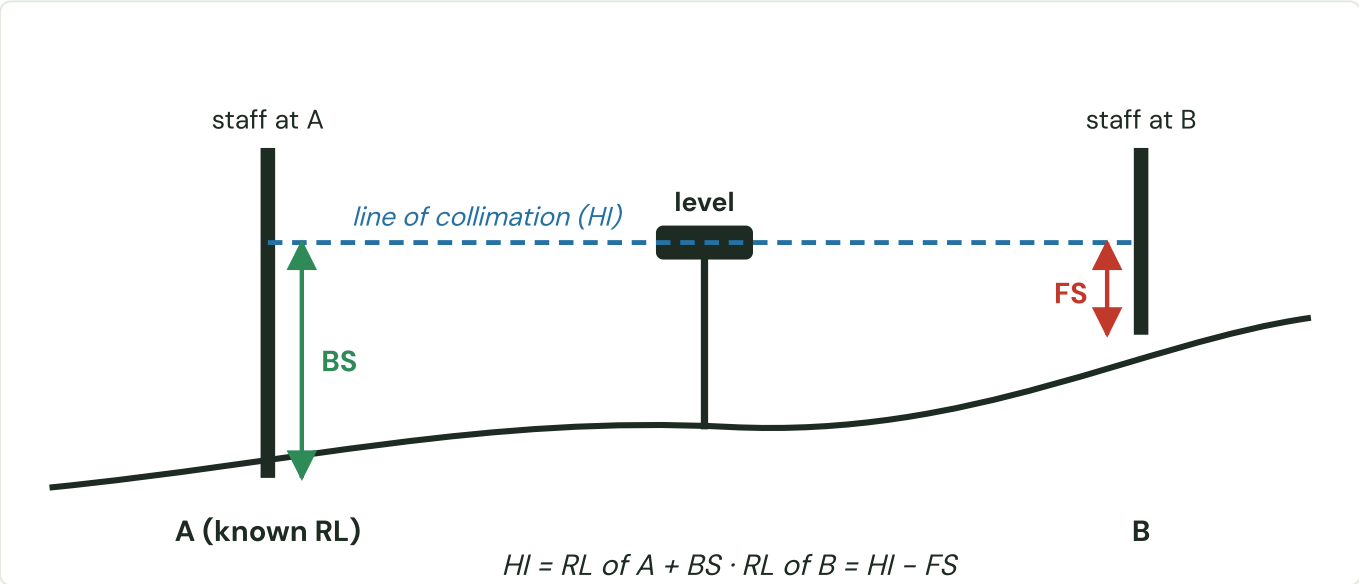
A bench mark (BM) is a point of known elevation from which other elevations are taken.

Type	Details
Permanent BM	Fixed by the national survey organisation — in India the Survey of India. Called G.T.S. (Grand Trigonometric Survey) bench marks; elevations refer to Mean Sea Level at Mumbai (Bombay). Found on bridge parapets, culverts, gate pillars, milestones and plinth corners of buildings
Arbitrary BM	Elevation assumed; for small levelling jobs
Temporary BM	Set when work breaks off, to continue later; placed on fairly permanent objects such as stones, pegs and gate posts

Methods and readings

- **Direct (spirit) levelling** — a line of sight perpendicular to gravity gives elevation differences between nearby points.
- **Trigonometric levelling** — computes the difference in elevation from the horizontal distance and the vertical angle.

Reading / point	Meaning
Back sight (BS), plus sight (+S)	First reading after setting up, taken on a point of known elevation; gives the HI
Fore sight (FS), minus sight (-S)	Last reading before shifting the instrument, on a point whose elevation is wanted
Intermediate sight (IS)	Any reading between the BS and FS from the same set-up
Turning point (TP) / change point	Point where the instrument is shifted; take both an FS and a BS on it; it must be firm so its elevation does not change
Height of instrument (HI)	RL of the line of sight — the height of the plane of collimation. With the instrument levelled, the line of sight sweeps a horizontal plane
Station	A point whose elevation is to be found — where the staff is held, not where the instrument stands



Back sight, fore sight and height of instrument

Levels

A level provides a horizontal line of sight. Choose the type by the accuracy you need.

Level	Features and use
Hand level	Small bubble tube and a telescope about 15 cm long, held upright; rough levelling for contour farming, terracing, bunding, strip cropping and estimating land slope
Abney hand level	A hand level with a graduated arc (degrees and per cent slope); measures vertical angles and slopes
A-frame level	Triangular wooden frame with a spirit level on the cross bar; lays out contours for bunds, terraces and conservation works by repeated positioning
Farm level	Low-cost spirit level with telescope on a portable tripod; when the bubble is centred the line of sight is level
Wye level	Telescope rests in two Y-shaped bearings and can be removed and reversed; the most expensive common level
Dumpy level	Telescope rigidly fixed to the frame; strong, stable and simple; the most commonly used. Cheaper than a wye level, fewer parts, less liable to go out of adjustment; suits ordinary levelling

Parts of a level

1. **Telescope** — metal tube with lenses, diaphragm and cross hairs; external- or internal-focusing.
2. **Objective** — forms the image.
3. **Eye-piece** — magnifies the image.
4. **Diaphragm** — brass ring with one vertical and three horizontal hairs: the middle hair gives the staff reading; the upper and lower (stadia) hairs give distance; the vertical hair checks that the staff is vertical.
5. **Bubble tube** — the line of sight is horizontal when the bubble is central.
6. **Levelling screws** — three or four; bring the bubble to the centre.
7. **Foot plate and tripod head** — support the instrument.

Size: stated by the focal length of the objective — commonly 30 cm and 35 cm; magnifying power 25 to 30.

Theodolite

- Its telescope revolves in both horizontal and vertical planes.
- **Uses:** horizontal and vertical angles, prolonging straight lines, stadia distances, levelling.
- **Size:** diameter of the horizontal circle, generally 10 cm to 25 cm. Telescope with stadia hairs, magnifying power 18 to 30. Horizontal and vertical circles with verniers reading to 20 seconds.

Levelling staves

Staff	Details
Self-reading (telescopic Sopwith)	Most used. Three telescoping lengths, each held by a brass spring catch; well-seasoned wood with metal cap and shoe; usually 4 m and 5 m long. Graduated in metres, decimetres and centimetres, metre figures in red on the left; read to two decimal places
Folding staff	Self-reading staff of two hinged parts; the upper part folds over the lower
Target rod	Two lengths sliding over each other, held by brass clamps; a movable target — usually a circular disc of about 12.5 cm diameter in red and white quadrants with a rectangular window — read by vernier to 0.5 mm

A simple telescope shows the staff upside down, because the eye-piece does not re-erect the image formed by the objective. Some staves carry inverted figures; some modern levels add an erecting lens.

Sources: ICAR eCourse 'Surveying and Leveling' — course content by Dr D. Tamil Mani (Dept. of Soil & Water Conservation Engg., AEC&RI, TNAU Coimbatore), reviewed by Dr V. M. Abdul Hakkim (KCAET, Kerala Agricultural University); published as an AgriMoon e-book. (Lessons 9–10) Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. A reading taken on a point of known elevation immediately after setting up is a

(a) fore sight
(b) back sight
(c) intermediate sight
(d) turning point

Q2. Height of instrument equals

(a) RL of BM − BS
(b) RL of BM + BS
(c) BS − FS
(d) RL + FS

Q3. Elevations of G.T.S. bench marks in India refer to MSL at

(a) Chennai
(b) Kolkata
(c) Mumbai
(d) Kandla

Q4. The level whose telescope is rigidly fixed to the frame is the

(a) wye level
(b) dumpy level
(c) Abney level
(d) hand level

Q5. The upper and lower horizontal hairs of the diaphragm are used for

(a) staff reading
(b) checking verticality
(c) stadia distance
(d) levelling the bubble

Q6. A telescopic Sopwith staff is usually

(a) 2 m or 3 m
(b) 4 m or 5 m
(c) 6 m or 8 m
(d) 10 m

Answers

Q1 — (b) back sight
Q2 — (b) RL of BM + BS
Q3 — (c) Mumbai
Q4 — (b) dumpy level
Q5 — (c) stadia distance
Q6 — (b) 4 m or 5 m